can you give me an example of an electric circuit with reistor, capacitor and inductor with an AC input and the associated equations that use complex numbers to solve.

Here's a clear example of an **RLC series circuit** driven by an AC source, solved using **complex numbers (phasors)**.

🔌 Example: Series RLC Circuit with AC Source

Circuit Components

- Resistor: R
- Inductor: L
- Capacitor: C
- AC voltage source:

$$v(t) = V_0 \cos(\omega t)$$

All components are connected **in series**.

1️⃣ Convert to Phasor (Complex) Form

The AC source:

$$v(t) = V_0 \cos(\omega t)$$

In phasor form:

$$\tilde{V} = V_0 \angle 0$$

2️⃣ Complex Impedances

Using complex numbers:

- Resistor:

$$Z_R = R$$

- Inductor:

$$Z_L = j\omega L$$

- Capacitor:

$$Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

Where:

- $j = \sqrt{-1}$
- $\omega = 2\pi f$

3️⃣ Total Impedance

Since components are in series:

$$Z_{total} = Z_R + Z_L + Z_C$$

$$Z = R + j\omega L - \frac{j}{\omega C}$$

Factor imaginary part:

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

4️⃣ Solve for Current (Using Ohm's Law in Complex Form)

$$\tilde{I} = \frac{\tilde{V}}{Z}$$

$$\tilde{I} = \frac{V_0}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

5️⃣ Convert to Polar Form

Magnitude of impedance:

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Phase angle:

$$\theta = \tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

So:

$$\tilde{I} = \frac{V_0}{|Z|}\angle(-\theta)$$

6️⃣ Time-Domain Current

Convert back:

$$i(t) = \frac{V_0}{|Z|}\cos(\omega t - \theta)$$

🔎 Numerical Example

Let:

- $R = 10\,\Omega$
- $L = 0.1\,H$
- $C = 100\,\mu F = 100 \times 10^{-6} F$
- $f = 50\,Hz$
- $V_0 = 100\,V$

Step 1: Angular Frequency

$$\omega = 2\pi f = 2\pi(50) = 314\,rad/s$$

Step 2: Reactances

Inductive:

$$\omega L = 314(0.1) = 31.4$$

Capacitive:

$$\frac{1}{\omega C} = \frac{1}{314 \times 100 \times 10^{-6}} \approx 31.8$$

Step 3: Total Impedance

$$Z = 10 + j(31.4 - 31.8)$$

$$Z = 10 - j0.4$$

Magnitude:

$$|Z| = \sqrt{10^2 + (-0.4)^2}$$

$$|Z| \approx 10.008$$

Phase:

$$\theta = \tan^{-1}(-0.4/10)$$

$$\theta \approx -2.29^\circ$$

Step 4: Current

Magnitude:

$$I_0 = \frac{100}{10.008} \approx 9.99\,A$$

Time domain:

$$i(t) = 9.99 \cos(314t + 2.29^\circ)$$

(Note: current leads slightly because circuit is slightly capacitive.)

⚡ Key Takeaways

- Replace components with **complex impedances**
- Use **Ohm's law with complex numbers**
- Convert back to time domain
- Phase angle tells you lead/lag behavior
- Resonance occurs when:

$$\omega L = \frac{1}{\omega C}$$

If you'd like, I can also show:

- The same circuit solved using differential equations
- A parallel RLC example
- A resonance analysis
- Or a phasor diagram visualization